Saint George International School

Design and Engineering

PORTFOLIO

By: JAF Racing Team





@jaf_racingteam

Driven by Innovation, United by Ambition

JAF RACING TEAM

JOURNEY • AMBITION • FOCUS

TESTING & PERFORMANCE ANALYSIS

START YOUR ENGINES!

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INITIAL OBJECTIVES

At the beginning, we focused on race performance, compliance, durability, and efficiency. Our goal was to create a fast, stable, and reliable car where every component contributed to overall performance.

- **RACE PERFORMANCE**

We focused on acceleration, stability, low resistance, aerodynamics, and lightweight design to improve racing performance.

- **COMPLIANCE**

All design decisions followed the competition regulations, including dimensions, materials, components, safety, and manufacturing requirements.

- **DURABILITY**

We evaluated the chassis, wheels, aerodynamic components, and connections to reduce unnecessary weight while maintaining strength and reliability.

LAUNCH EFFICIENCY

Our testing showed us that the launch has a major influence on the final race time. We therefore studied the relationship between car mass, wheel design, bearing friction, and rotational inertia.

After comparing our results, we focused on reducing unnecessary resistance and rotational mass so that more of the available energy could be converted into forward movement.

TRACK TESTING

We used track testing to determine whether the improvements we made in CAD and analysis actually translated into better physical performance.

We performed repeated runs and compared acceleration, stability, wheel behavior, trajectory, and race consistency. The results allowed us to identify weaknesses and make further changes to the design.

WHAT WE OBSERVED

- Changes in acceleration after reducing mass.
- Differences in wheel performance.
- Effects of alignment and stability.
- Aerodynamic improvements between designs.
- Consistency between repeated runs.

DESIGN ANALYSIS

We analyzed our car's design to identify areas that could be improved before and during testing. We focused on mass, aerodynamics, wheel performance, stability, and durability. After each test, we compared the results with our expectations, identified weaknesses, and made adjustments to improve the overall performance of the car.

1. Mass — We reduced unnecessary weight while maintaining structural strength.

2. Aerodynamics — We evaluated the shape of the car to reduce drag and improve airflow.

3. Wheel Performance — We focused on wheel alignment, friction, and smooth rotation.

4. Stability — We adjusted the design to maintain a straight and controlled trajectory.

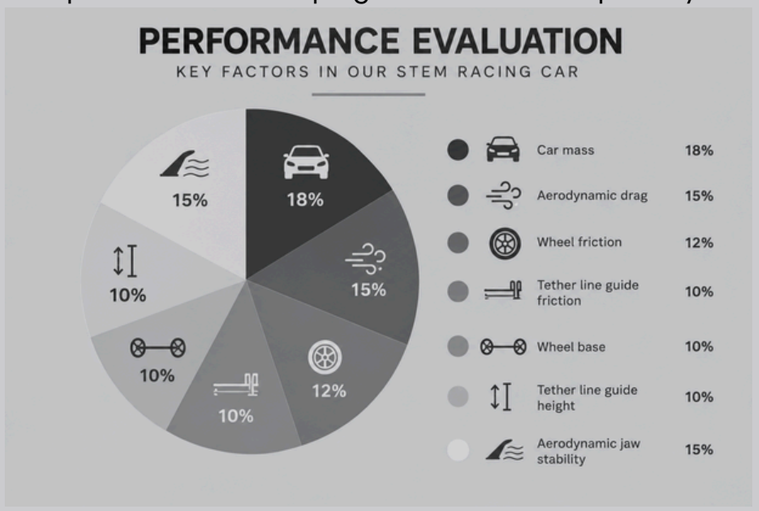
5. Durability — We tested the components to make sure they could withstand repeated racing conditions.

6. Design Improvements — We used our test results to make changes and progressively improve the final car.

KEY FINDINGS

After analyzing our designs and conducting tests with the actual car, we discovered that no single modification was solely responsible for the improved performance.

Our best results were achieved by combining a lightweight structure, functional wheels, reduced friction, controlled aerodynamics, and good stability. This demonstrated to us the importance of developing the car as a complete system.



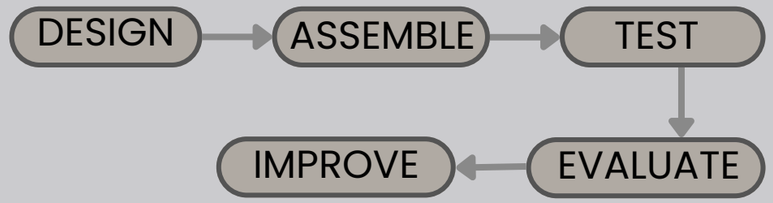
THRUST EFFICIENCY

TIPPING MOMENT

During our analysis, we also considered the possibility of the car becoming unstable during racing.

We evaluated the relationship between the center of mass, car geometry, and forces acting on the vehicle. Our testing and analysis helped us maintain a stable design without adding unnecessary mass.

METHODOLOGY



AERODYNAMIC DESIGN RESEARCH

AERODYNAMIC EFFICIENCY

Our vehicle's aerodynamic design focuses on controlling airflow around the front, sides, and rear. We analyzed the geometry of each aerodynamic component to reduce unnecessary drag while maintaining stability.

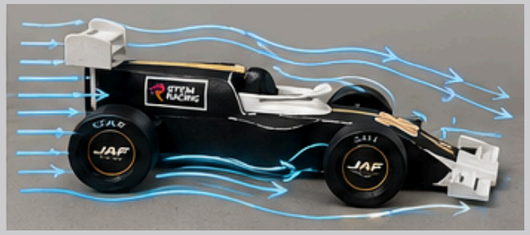


Figure 1. Expected aerodynamics for the car body.

DESIGN OBJECTIVE:

- Redirect incoming airflow.
- Improve front-end stability.
- Reduce airflow interference around the wheels.
- Establish controlled airflow toward the rest of the vehicle.

AERODYNAMIC EFFICIENCY

Our vehicle's aerodynamic design focuses on controlling airflow around the front, sides, and rear. We analyzed the geometry of each aerodynamic component to reduce unnecessary drag while maintaining stability.

DESIGN OBJECTIVE:

- The rear wing was analyzed to find an effective balance between aerodynamic stability and drag. Additionally, its supports and geometry were considered as part of the overall aerodynamic system.

FRONT WING

The front wing is the primary aerodynamic component interacting with the incoming airflow. Its geometry influences how air is directed toward the front wheels, the underbody, and the sidepods.

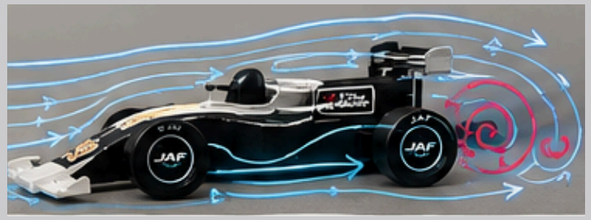


Figure 2. Vortex and forward flow expected from the front wing.

RESEARCH APPROACH

- We compared different sidepod geometries to determine how their shape could improve airflow management while maintaining a compact and manufacturable design.

MULTI-ELEMENT WINGS

- Multi-element wings utilize multiple aerodynamic surfaces working in concert, rather than relying on a single surface. This configuration allows for progressive airflow management across the wing.

APPLICATION IN OUR VEHICLE

- The use of multiple aerodynamic elements was considered to improve airflow control while maintaining a compact, functional design.

WHEEL WAKE

Our vehicle's aerodynamic design focuses on controlling airflow around the front, sides, and rear. We analyzed the geometry of each aerodynamic component to reduce unnecessary drag while maintaining stability.



Figure 3. View of the rear spoiler

DESIGN OBJECTIVE:

- Our aerodynamic development accounts for the positioning of the front wing and side components to control the airflow generated around the wheels.

CENTER OF MASS AND STABILITY

- The position of the center of mass influences vehicle stability during acceleration, movement, and changes in direction.
- In our design, maintaining a balanced mass distribution is crucial to avoid unwanted instability and ensure a predictable trajectory.

DESIGN OBJECTIVE:

- Maintain a balanced, low mass distribution by properly integrating aerodynamic components and structural elements.

SIDEPOD AIRFLOW

Sidepods channel airflow from the front section toward the rear of the vehicle. Their shape influences the smoothness of the airflow transition along the bodywork.



Figure 4. Smooth transition between the body and the parts.

DESIGN OBJECTIVE:

- Achieve a smoother airflow transition.
- Control airflow toward the rear. Reduce unnecessary turbulence.
- Integrate the side pods with the front and rear aerodynamic systems.

WHEEL PERFORMANCE

- Wheels interact directly with the track; therefore, their geometry and alignment are crucial to the vehicle's movement.

RESEARCH FOCUS

ASPECT	DESCRIPTION
WHEEL ALIGNMENT	Ensures the wheels are properly aligned with the track to maintain stability and direction.
ROLLING RESISTANCE	Minimizes energy loss between the tire and the track, allowing for smoother movement.
WHEEL ROTATION	Allows the wheels to spin freely and consistently, reducing friction and improving traction.
WHEEL MASS	Keeping the wheels as light as possible improves acceleration and overall performance.
TRACK CONTACT	Ensures maximum grip and proper transfer of power between the car and the track.
DESIGN OBJECTIVE	Minimize unnecessary resistance while maintaining precise alignment and smooth rotation.

CAR DEVELOPMENT

STARTING POINT

This is the first time our team has participated in a STEM Racing competition, so at the beginning, we were unsure how many modifications we were allowed to make to the car compared to the standard design. To clarify this, we began by reviewing the competition regulations to identify the required dimensions, components, and materials, as well as the parts we were allowed to customize. These included the front and rear wings, the shape of the car's body, which could be freely designed as long as it did not touch the No-Go Zone, and the minimum weight requirement, among other specifications.

INITIAL SKETCH

Once we understood these limitations, we spent several days discussing and proposing ideas. Each team member contributed concepts regarding the body shape, aerodynamics, and visual appearance of the car. We selected the final sketch through a vote, choosing the design that was the most practical and best aligned with our shared vision as a team.

After reaching a consensus, we transferred the winning idea onto a hand-drawn sketch. This drawing served as the foundation for visualizing the car's overall proportions and as a guide for the subsequent design process.

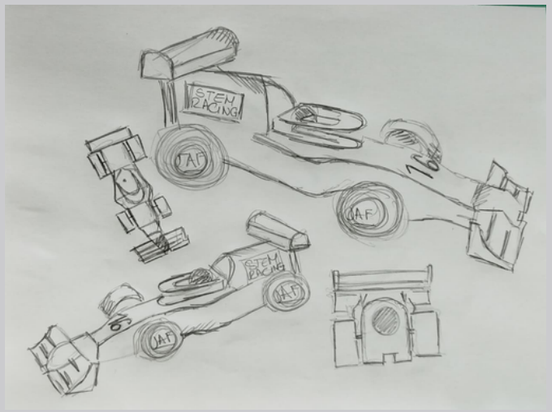


Figure 5. Initial freehand sketch, which served as the starting point for the design process.

FROM PAPER TO DIGITAL MODEL

Once the sketch was finalized, the next step was to transfer the concept into a digital environment. We used **Autodesk Fusion 360** to create a 3D model of the car. During this process:

- We adapted the original design to meet the mandatory dimensions and specifications established by the competition.
- We adjusted the shape of the car to balance our design preferences with the requirements of the competition regulations.
- We maintained our main design ideas, including the distribution and shape of the body, while ensuring that the design complied with the limits of the No-Go Zone.
- The No-Go Zone did not significantly restrict our aerodynamic design choices for the front nose. We also designed the Halo to be as smooth as possible to minimize its impact on the car's overall aerodynamics.

Parameter	Name	Unit	Expression	Value	Comments
User Parameter	H_bloque	mm	50 mm	50.00	Apéndice i
User Parameter	D_cartucho	mm	18.0 mm	18.00	Apéndice ii – verificado
User Parameter	P_cartucho	mm	52.0 mm	52.00	Apéndice ii – verificado
User Parameter	H_eje_cart	mm	29.0 mm	29.00	Apéndice ii – eje sobre t
User Parameter	D_rueda	mm	30.0 mm	30.00	Apéndice v
User Parameter	W_rueda	mm	16.5 mm	16.50	Apéndice v
User Parameter	D_eje	mm	3.0 mm	3.00	Apéndice vi
User Parameter	D_guia_eje	mm	6.35 mm	6.35	Apéndice iii
User Parameter	H_eje_rueda	mm	11.0 mm	11.00	D4.3.3 – ancla del Halo

Figure 6. Standard dimensions and specifications for STEM Racing.

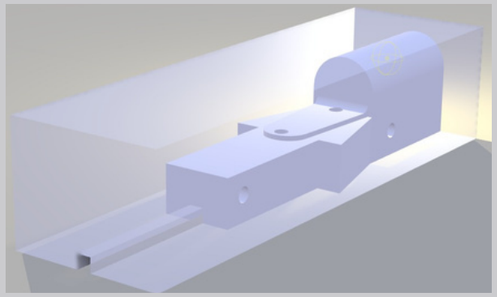


Figure 7. No-Go Zone, the restricted area that must remain untouched to prevent structural weakening and potential safety risks.

Once all the restrictions had been defined and the standard components, including the wheels, hubs, axles, and Halo, had been accounted for, we began modeling the car according to our own design preferences.

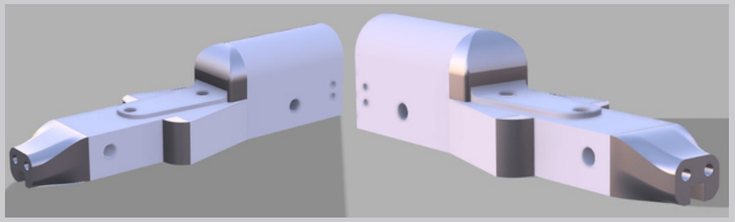


Figure 8. Car body modeled in Fusion 360 and adapted to our original concept.

Finally, we added the front and rear wings, completing the model and preparing it for further testing, including strength, drag, and airflow analysis.

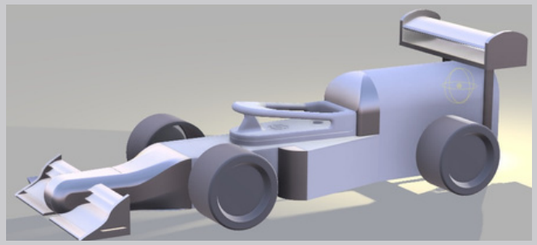


Figure 9. Final digital model showing how the car is expected to look once manufactured.

OUR DESIGN CONCEPT

Our design aims to address two aerodynamic challenges simultaneously. The first is to generate **downforce.** To achieve this, the car's aerodynamic surfaces, including the wings and the contours of the body, are designed to direct airflow in a way that generates a downward force. According to Newton's third law of motion, changes in the direction of airflow produce corresponding forces on the car.

This helps increase its grip and stability on the track. It is particularly important for meeting the competition requirement that all four wheels remain in contact with the track throughout the race, reducing the risk of the car lifting off or becoming unstable at high speeds.

The second challenge is to minimize **air resistance (drag).** For this reason, we aimed to create smooth, streamlined, and tapered shapes that allow air to flow around the car with minimal turbulence, reducing resistance and allowing it to move more efficiently.

CAR DEVELOPMENT

To support this objective, we designed a low front nose and incorporated a wing that helps control and redirect airflow as needed.

Although both objectives are important, they can conflict with each other because generating greater downforce often increases drag. Therefore, our design aims to achieve the best possible balance between stability and speed. This balance can be observed in the wind tunnel test **(Figure 10.)**, where the airflow around the car appears balanced and controlled.

AERODYNAMIC TESTING

FORMULA 1 CAR – AERODYNAMIC CFD ANALYSIS

The images show a Computational Fluid Dynamics (CFD) simulation of a Formula-style racing car. The purpose of this analysis is to understand airflow behavior, identify potential turbulence zones, and evaluate factors affecting aerodynamic drag, downforce, and vehicle stability.

INTERPRETATION OF THE CFD RESULTS

The simulation uses colored streamlines to illustrate airflow around the vehicle. Since the original images do not include a numerical color scale, the following interpretations are preliminary.

Table 1. These colors may represent velocity or another simulation variable, depending on the visualization settings.

COLOR	POSSIBLE INTERPRETATION
Yellow	Relatively fast, uniform airflow
Green	Airflow deflection around the body
Cyan	Potentially slower or redirected airflow
Blue	Possible low-speed regions, vortices, or wake structures

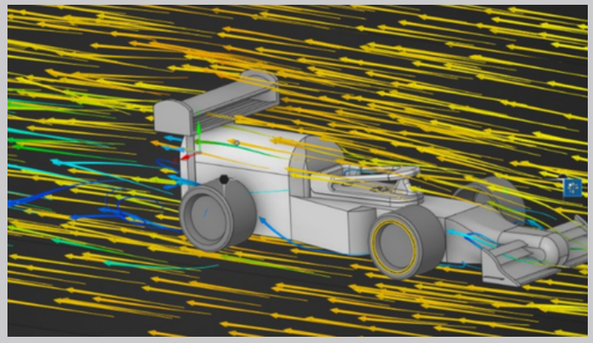


Figure 10. Digital study using Ansys Fluent on the vehicle's actual aerodynamics.

AERODYNAMIC OBSERVATIONS

Front section: Airflow separates around the nose, front wing, and front wheels. The wheels create flow disturbances that may increase aerodynamic drag.

Central section: Air travels around the cockpit and side bodywork. The geometry redirects airflow toward the upper and lower surfaces of the car.

Rear section: Curved streamlines suggest wake formation and possible recirculation behind the rear wheels and rear wing.

Rear wing: The wing can generate downforce, improving tire grip and stability, but it also produces aerodynamic drag.

Underbody: The floor and diffuser can contribute significantly to downforce. However, their performance cannot be determined from the provided streamline images alone.

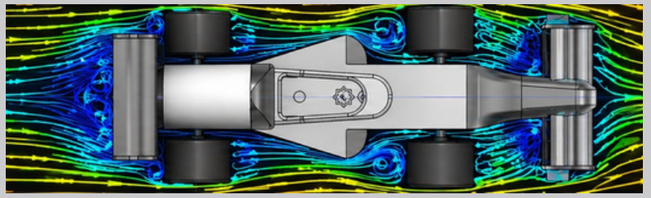


Figure 11. Dynamic flow of the car body, viewed from above.

AERODYNAMIC PERFORMANCE EVALUATION

Table 2. The table summarizes airflow behavior and turbulence, highlighting the need for numerical data to assess aerodynamic efficiency.

PARAMETER	QUALITATIVE EVALUATION
Airflow over bodywork	Relatively uniform in several regions
Front wheel turbulence	Visible airflow disturbances
Rear wake	Curved streamlines and possible vortices
Aerodynamic drag	Present but not quantified
Downforce generation	Requires pressure and force data
Overall efficiency	Cannot be confirmed without numerical results

RECOMMENDATIONS FOR IMPROVEMENT

1. **Front wing optimization:** Adjust the angle of attack to improve airflow management and evaluate downforce generation.

2. **Wheel turbulence reduction:** Investigate airflow separation around the front and rear wheels.

3. **Underbody optimization:** Improve floor and diffuser geometry to increase downforce while minimizing drag.

4. **Rear wing analysis:** Evaluate the relationship between rear-wing downforce and aerodynamic resistance.

5. **Numerical validation:** Calculate the drag coefficient (\(C_d\)), lift coefficient (\(C_l\)), and total downforce under defined simulation conditions.

DESIGN EVALUATION

The aerodynamic simulation provides valuable insight into how a Formula-style racing car interacts with surrounding airflow. The streamlines reveal relatively uniform flow away from the vehicle, while disturbances appear around the wheels, bodywork, and rear section. These regions are particularly important because they can influence aerodynamic resistance and vehicle stability.

The results highlight potential areas for aerodynamic improvement, especially around the front wing, underbody, rear wheels, and rear wing. However, numerical pressure distributions, velocity measurements, and aerodynamic force calculations are necessary to determine the actual performance of the design.

Overall, the CFD visualization is a useful starting point for evaluating and improving the aerodynamic characteristics of the racing car, but further quantitative analysis is required to validate its efficiency.

COMPUTER-AIDED MANUFACTURING

CAM

The Computer-Aided Manufacturing (CAM) stage focused on preparing selected components of our design for physical production. For this part of the project, we concentrated mainly on the front and rear wings, which were important elements of the car's appearance and aerodynamic design. After developing the digital designs, we used Creality Print to prepare the components for 3D printing.

Creality Print allowed us to prepare the digital files and organize the printing process before producing the physical pieces. During this stage, we reviewed the positioning of the components and their preparation for printing to ensure that the files were suitable for the process. This step was important because the quality of the printed parts depended not only on the original design but also on how the components were prepared for production.

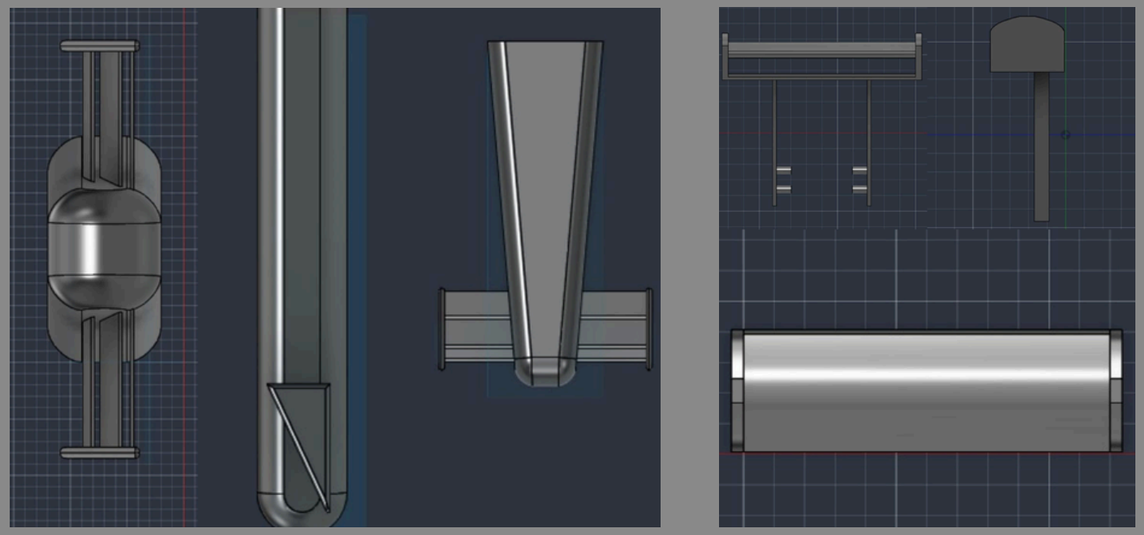


Figure 13. Front and rear wing files prepared in Creality Print and ready for 3D printing.

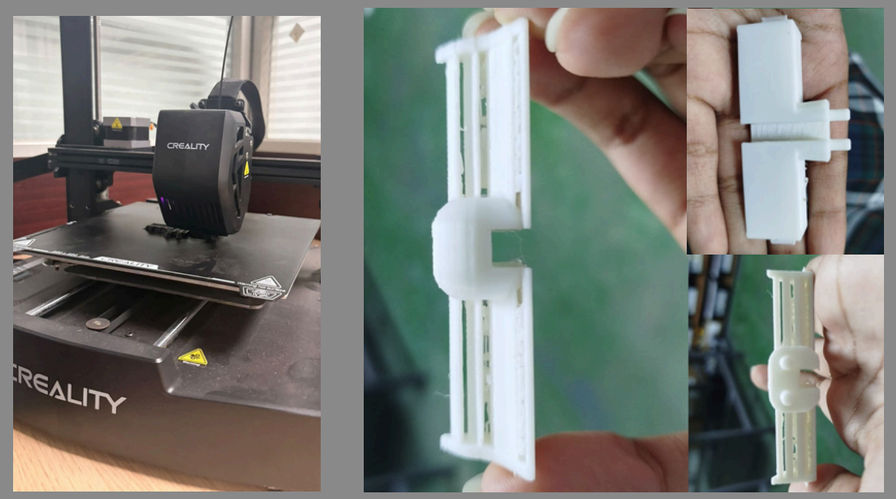


Figure 14. 3D-printed front wing prepared for the final assembly of the JAF Racing Team car.

After preparing the files, we printed the wings and evaluated the physical results. We examined the components to determine whether they had the required shape, whether their structure appeared sufficiently resistant, and whether they could withstand normal handling and assembly. We also considered their durability because the parts needed to remain in good condition throughout the preparation and testing of the car.

The first printed versions helped us understand how our digital designs translated into physical components. By examining the printed pieces, we could identify areas that needed adjustments and determine whether the designs met our expectations. This connection between digital preparation and physical evaluation allowed us to improve the components before settling on their final versions.

In addition to the wings, we also used 3D printing to produce our own halos for the final car. Although some components were initially provided to us, preparing and printing our own halos allowed us to incorporate these pieces into the final design. Overall, the CAM stage helped us turn our digital designs into physical parts and evaluate their quality, resistance, and suitability for the completed racing car.

DESIGN PROCESS EVALUATION

CAD (COMPUTER-AIDED DESIGN)

We used **Fusion 360** to create a digital model of our JAF Racing Team car, developing its structure, shape, dimensions, and main components. This allowed us to visualize how the parts would fit together, including the front and rear wings, and make adjustments before preparing them for printing. Throughout the process, we focused on functionality, stability, and incorporating our team's red, gold, and black identity. The digital model helped us identify possible improvements, organize our ideas, and establish a solid foundation for the car's final design.



Figure 12. Physical car body being painted and prepared for final assembly.

TESTING & EVALUATION

Testing was an essential part of evaluating the performance of our JAF Racing Team car. Once the main components had been assembled, we tested the vehicle on the track to observe how it performed under the conditions available to us. Our main objective was to evaluate its movement, speed, and stability and determine whether the final configuration met our expectations.

During the tests, we paid attention to how the car moved along the track and whether it maintained stable movement during its run. We also observed the general behavior of the vehicle to determine whether the assembled components appeared to function together as intended. These observations allowed us to evaluate the result of the work completed during the design, printing, and assembly stages.

Testing was particularly useful because it allowed us to assess the physical car rather than relying only on its digital appearance. A design that looks appropriate on a computer must still be evaluated once its components have been produced and assembled. By testing the actual vehicle, we could observe its performance and determine whether further modifications were necessary.

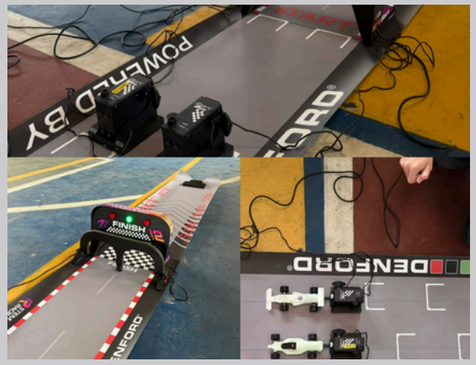


Figure 15. Final track test of the completed JAF Racing Team car to evaluate its speed and stability.

After reviewing the results of our tests, we were satisfied with the speed and stability we observed. We therefore decided to maintain the overall configuration of the car instead of making additional changes to its main structure. This decision was based on our evaluation of the performance achieved during the tests and our satisfaction with the results.

Although we did not introduce further changes to the overall configuration after testing, the evaluation stage remained important because it gave us the opportunity to confirm that the car behaved as expected. It also helped us develop a better understanding of the relationship between the design choices we made and the performance of the completed vehicle.

DESIGN IMPROVEMENTS

We refined the front and rear wings by reviewing the first 3D-printed versions and identifying areas that could be improved. We adjusted their designs to achieve more suitable shapes, better integration with the car, and adequate resistance for handling and assembly.

Comparing the digital models with the physical components helped us identify necessary modifications and improve the final pieces. These adjustments were part of our iterative design process, while the car's overall configuration remained unchanged after testing because we were satisfied with its speed and stability.

FINAL DESIGN REVIEW

The final review focused on evaluating our JAF Racing Team car as a complete project. We checked the overall appearance, assembly quality, positioning of the components, and condition of the printed wings and halos. We also reviewed the stickers and color palette to ensure the car reflected our team's visual identity. Since we were satisfied with its speed and stability during track testing, we maintained the overall configuration without further structural changes. Finally, we considered the competition requirements and the objectives established by our team. This review helped us recognize the importance of combining digital design, component preparation, assembly, testing, and teamwork to achieve the final result.



Figure 16. Completed JAF Racing Team car, fully assembled and ready for delivery and competition.

ACKNOWLEDGEMENTS

Being part of the **STEM Racing in Panama** project for the first time has been an unforgettable experience for our entire team. Throughout this journey, we had the opportunity to learn new skills, explore our creativity, apply our knowledge of science, technology, engineering, and mathematics, and discover what it truly means to work together toward a common goal.

From designing our car and developing our ideas to manufacturing components, testing our prototype, and improving our design, every stage brought new challenges, lessons, and moments of excitement. Although some parts of the process were challenging, overcoming them as a team made the experience even more rewarding. We are especially grateful to **Saint George International School of Panama** for giving us the opportunity to participate in this project and for supporting us throughout our journey.
We would also like to express our sincere appreciation to our teachers, mentors, and everyone who guided us, shared their knowledge, and encouraged us to keep moving forward.
A special thank-you goes to **STEM Racing** for creating a platform that allows students to experience engineering and innovation in such a practical, creative, and exciting way.

This project was something new for us, and it allowed us to step outside our comfort zone, develop confidence, strengthen our teamwork, and create memories that we will carry with us beyond the competition.
We feel proud of what we have accomplished as the **JAF Racing Team**, not only because of the car we designed, but also because of everything we learned and experienced along the way.
We finish this journey with gratitude, excitement, and a greater appreciation for engineering, collaboration, and innovation. This is more than just a racing project; it is a reminder that with ambition, dedication, and teamwork, we can turn our ideas into reality. Thank you for being part of our journey.